

MAINCRAFTS TECHNOLOGY

Cybersecurity Internship Program

CYBERSECURITY TASK-1

Threat Intelligence Report

Awareness & Research Project

Submitted by

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1. Introduction to Cybersecurity

1.1 What is Cybersecurity?

Cybersecurity is the practice of protecting computers, servers, networks, mobile devices, and the data they hold from unauthorized access, damage, disruption, or theft. It combines technologies, processes, and people-focused practices such as access control, encryption, monitoring, and awareness training to defend the confidentiality, integrity, and availability of information systems. As individuals and organizations move more of their lives and operations online, cybersecurity has evolved from a purely technical concern handled by IT departments into a core business and personal-safety function.

1.2 Why It Matters for Individuals and Businesses

For individuals, cybersecurity protects personal identity, banking details, health records, and private communications from theft or misuse. A single compromised account can lead to financial fraud, identity theft, or reputational harm. For businesses, the stakes are even higher: a breach can halt operations, expose customer data, trigger regulatory penalties, and permanently damage brand trust. Critical sectors such as healthcare, finance, energy, and government depend on secure systems to keep essential services running, which means a cyberattack on any one of them can have consequences that ripple across society.

1.3 Current Relevance

Cybercrime has continued to escalate through 2024 and 2025. Ransomware-as-a-Service platforms have lowered the technical barrier for launching attacks, generative AI is being weaponized to craft convincing phishing emails and deepfake impersonations, and the rapid, sometimes rushed migration of workloads to the cloud has left many organizations exposed through simple misconfigurations. At the same time, the number of internet-connected IoT devices keeps growing, expanding the attack surface, while unpatched zero-day vulnerabilities in widely used enterprise software continue to be exploited within days of disclosure. Together, these trends make continuous vigilance, timely patching, and security awareness more important than ever.

2. Major Modern Cyber Threats, Impact, Case Studies & Preventive Measures

The five threats below were selected because they represent the categories responsible for the largest share of reported incidents, financial losses, and regulatory attention in 2024-2025. Each threat is analyzed with its impact on individuals and organizations, a real-world incident, and recommended defense measures.

2.1. AI-Powered Phishing Attacks (including Deepfakes)

Attackers now use generative AI to write highly convincing phishing emails, clone voices from short audio samples, and generate realistic video of executives in real time. This removes the spelling mistakes and awkward phrasing that once made phishing easy to spot, and allows scams to be personalized at scale using information scraped from social media and public records.

Impact Analysis

- **Individuals:** Individuals can be tricked into revealing passwords, OTPs, or banking credentials, or into believing a fraudulent call or video from a "relative" or "official" is genuine, leading to direct financial loss.
- **Organizations:** Organizations face large-scale wire-transfer fraud, business email compromise, and credential theft that can bypass traditional security-awareness training, since employees are conditioned to trust a familiar voice or face.

Real-World Case Study

In January 2024, a finance employee at the Hong Kong office of British engineering firm Arup joined a video conference believing he was speaking with the company's CFO and several colleagues. Every participant except the employee was an AI-generated deepfake, created from publicly available video and audio of the real executives. Convinced by the video call, the employee authorized 15 transactions worth roughly USD 25 million to accounts controlled by the attackers.

Source: Reported by The Guardian and multiple cybersecurity outlets (Arup deepfake fraud case, 2024)

Preventive Measures

- Mandatory out-of-band verification (a phone call to a known number) for any request involving fund transfers or credential changes, even if approved on a video call.
- Deploy AI-aware email and voice-authentication security tools that flag synthetic media indicators.
- Run regular, realistic phishing and deepfake-simulation training so employees learn to question unusual urgency, not just spelling errors.

2.2. Ransomware-as-a-Service (RaaS)

Ransomware-as-a-Service lets criminal groups "rent" ready-made ransomware kits, infrastructure, and even customer support to affiliates who carry out attacks in exchange for a share of the ransom. This business model has professionalized ransomware, allowing less technically skilled criminals to launch sophisticated, double-extortion attacks that both encrypt data and threaten to leak it publicly.

Impact Analysis

- **Individuals:** Individuals whose personal data is held by an affected organization can have sensitive records (medical, financial, or identity information) exposed or sold on the dark web.

- Organizations: Organizations suffer prolonged downtime, costly recovery, regulatory fines, lawsuits, and lasting reputational damage; many are pressured into paying multi-million-dollar ransoms with no guarantee that stolen data will not still be leaked.

Real-World Case Study

The February 2024 ransomware attack on Change Healthcare, a UnitedHealth Group subsidiary that processes a large share of U.S. medical claims, disrupted pharmacy and billing systems nationwide for weeks. Linked to the ALPHV/BlackCat RaaS operation, the incident compromised the protected health information of over 100 million individuals and is estimated to have cost the company more than USD 2.4 billion, making it one of the costliest healthcare breaches on record.

Source: Widely reported by HHS, UnitedHealth Group disclosures, and industry ransomware trackers (2024)

Preventive Measures

- Maintain offline, regularly tested backups following the 3-2-1 backup rule to enable recovery without paying a ransom.
- Apply a Zero Trust security model with network segmentation to limit how far an attacker can move after an initial breach.
- Enforce timely patch management and Multi-Factor Authentication (MFA) on all remote-access and administrative accounts.

2.3. Cloud Security Misconfigurations

As organizations move data and applications to platforms like AWS, Azure, and Google Cloud, simple configuration mistakes such as publicly accessible storage buckets, overly permissive access controls, or hard-coded credentials in source code have become one of the leading causes of large-scale data exposure. Under the cloud "shared responsibility model," the provider secures the underlying infrastructure, but the customer is responsible for configuring it correctly.

Impact Analysis

- Individuals: Individuals whose personal or payment data is stored in a misconfigured cloud service can have that information exposed to anyone who finds the open resource, often without any "hacking" being required at all.
- Organizations: Organizations face data breaches that trigger regulatory investigations, compliance violations (e.g., GDPR, HIPAA), loss of customer trust, and in some cases direct financial exploitation of exposed cloud credentials.

Real-World Case Study

In 2024, researchers discovered that Football Australia had left AWS S3 storage buckets misconfigured, with access keys embedded in plain text in the organization's website source code. This exposed 127 digital storage containers containing players' personal details, ticket-purchase data, and internal source code, illustrating how a single coding oversight can cascade into a major exposure of sensitive information.

Source: Cloud Security Alliance, Top Threats to Cloud Computing Deep Dive (2025)

Preventive Measures

- Enforce least-privilege Identity and Access Management (IAM) policies and routinely audit permissions for over-privileged roles.
- Block public access on storage buckets by default and enable continuous configuration monitoring (CSPM tools) to catch drift.
- Never hard-code credentials in source code; use secrets-management services and rotate keys regularly.

2.4. IoT Vulnerabilities

Internet of Things (IoT) devices, ranging from home routers and IP cameras to smart appliances and industrial sensors, are often shipped with weak default credentials, infrequent security updates, and limited built-in protection. Attackers exploit these weaknesses to recruit devices into large botnets used for massive Distributed Denial-of-Service (DDoS) attacks or as entry points into home and corporate networks.

Impact Analysis

- **Individuals:** Individuals face privacy violations (an unsecured camera or smart-home hub can be hijacked to spy on them) and can have their home internet connection unknowingly used to attack others.
- **Organizations:** Organizations relying on IoT sensors in operations (e.g., manufacturing, logistics, healthcare) risk operational disruption, and their networks can be knocked offline entirely if targeted by an IoT-botnet-driven DDoS attack.

Real-World Case Study

The Aisuru botnet, first identified in late 2024, compromised an estimated 300,000 to 700,000 routers, DVRs, and IP cameras by exploiting weak credentials and unpatched firmware. By late 2025 it was capable of launching DDoS attacks exceeding 29.7 terabits per second, among the largest ever recorded, and its operators also rented out the hijacked devices as criminal proxy infrastructure.

Source: Cloudflare, Krebs on Security, and U.S. Department of Justice disruption reports (2025-2026)

Preventive Measures

- Change default device credentials immediately and disable unused remote-access features on routers and smart devices.
- Keep IoT firmware updated and isolate IoT devices on a separate network segment from critical business or personal systems.
- Use network-level monitoring or a firewall/IDS to detect unusual outbound traffic that may indicate a device has been compromised.

2.5. Zero-Day Exploits

A zero-day vulnerability is a security flaw that is exploited by attackers before the software vendor has released a patch, leaving "zero days" of warning for defenders. These are especially dangerous when found in widely used enterprise software, since a single flaw can be used to compromise thousands of organizations before a fix is even available.

Impact Analysis

- **Individuals:** Individuals are rarely targeted directly but are frequently affected indirectly when a zero-day is used to breach an organization holding their personal data.
- **Organizations:** Organizations can suffer a rapid, large-scale compromise; because no patch exists at the time of the attack, traditional patch-management defenses are ineffective until the vendor responds, leaving a costly window of exposure.

Real-World Case Study

Beginning in December 2024, a suspected nation-state threat group exploited a zero-day vulnerability (CVE-2025-0282) in Ivanti Connect Secure VPN appliances to gain unauthenticated remote code execution on victim networks. This followed a nearly identical pair of Ivanti zero-days exploited exactly a year earlier, showing how the same class of edge-network products can be repeatedly targeted, forcing CISA to issue emergency directives requiring U.S. federal agencies to disconnect or patch affected devices immediately.

Source: Mandiant/Google Cloud Threat Intelligence and CISA advisories (2024-2025)

Preventive Measures

- Apply vendor patches and security advisories as a priority, with a documented emergency-patching process for internet-facing systems.
- Deploy Intrusion Detection/Prevention Systems (IDS/IPS) and endpoint detection tools that can flag anomalous behavior even without a known signature.
- Reduce the attack surface by limiting internet exposure of administrative interfaces (VPNs, file-transfer tools) and monitoring vendor Known-Exploited-Vulnerabilities lists.

3. Threat Summary at a Glance

Threat	Primary Risk	2024-2025 Case Study	Key Defense
AI-Powered Phishing	Fraudulent fund transfers, credential theft	Arup deepfake fraud, ~USD 25M lost	Out-of-band verification
Ransomware-as-a-Service	Data encryption & extortion	Change Healthcare breach, 100M+ records	Offline backups, MFA
Cloud Misconfiguration	Unauthorized data exposure	Football Australia S3 bucket leak	Least-privilege IAM, CSPM
IoT Vulnerabilities	Botnets, DDoS, spying	Aisuru botnet, 29.7 Tbps DDoS	Change defaults, network isolation
Zero-Day Exploits	Unpatched remote code execution	Ivanti Connect Secure CVE-2025-0282	Emergency patching, IDS/IPS

4. Conclusion & Future Scope

The threat landscape reviewed in this report shows that modern cyberattacks are no longer limited to a single technique or target. Attackers now combine artificial intelligence, cloud-native tactics, and mass-scale automation to compromise individuals and organizations of every size, from small businesses to national healthcare systems. What makes 2024-2025 distinct is the speed and accessibility of these attacks: Ransomware-as-a-Service has turned ransomware into a commodity, generative AI has made social engineering nearly indistinguishable from genuine communication, and the sheer scale of cloud infrastructure and IoT devices has multiplied the number of possible entry points.

Proactive cybersecurity is therefore no longer optional. Organizations that combine strong technical controls (patch management, Zero Trust architecture, MFA, encryption) with continuous employee awareness training are far better positioned to detect and contain incidents before they become catastrophic. For individuals, basic hygiene such as using unique passwords, enabling MFA, and staying skeptical of urgent requests remains the first line of defense.

Looking ahead, threats will keep evolving alongside the technologies meant to defend against them: AI will be used both to attack and to defend, quantum computing may eventually challenge current encryption standards, and the growth of IoT and cloud adoption will continue to expand the attack surface. This makes continuous learning essential for any cybersecurity professional. Completing tasks like this one, tracking evolving threats, studying real incidents, and practicing structured analysis, builds the habit of staying current that the field demands.

5. References

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